

Interoperability Maturity Lab (IML)

IML Database and Country Evaluation Engine

Technical Manuscript for Evidence-Guided, Human-Validated Country Profiles

Companion technical document to the IML Founding Manuscript

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Health is the objective. Trustworthy information is the foundation. Interoperability is the path.

The engine organises evidence and proposes a reproducible starting point. Human review remains decisive.

Document Status and Reading Guide

Working technical document

This manuscript documents the current IML database, publication API and country evaluation logic, together with the audit components under consolidation. It is not a final software specification, a validated country ranking or a substitute for expert country review.

The IML Founding Manuscript establishes the scientific and conceptual framework. This technical manuscript explains how country information can be stored, audited, interpreted, scored, reviewed and published through a versioned PostgreSQL-based engine. The two documents are complementary: the Founding Manuscript defines why and what IML assesses; the Technical Manuscript describes how the current prototype organises evidence and supports human validation.

The architecture described here reflects the project state in July 2026. Some components are already operational in the public application, while others remain experimental or are being consolidated. Every section therefore distinguishes implemented elements from target design.

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Executive Summary

The IML country evaluation engine is designed as an evidence-guided research instrument. It collects structured country information, connects documentary sources to IML indicators, records uncertainty and limitations, proposes reproducible country profiles, and exposes validated or explicitly provisional results to the public website. Its purpose is not to replace country expertise. Its purpose is to make the starting point for country research visible, auditable and correctable.

The current implementation uses PostgreSQL hosted on Neon, a serverless API deployed on Vercel, and a React/Vite public interface. The database stores country metadata, profile versions, six domain scores, strengths, points to watch, documentary sources, indicator-level evidence links and assessment records. The API publishes only active countries with published profiles and returns the information required by the World Map and Country Report interfaces.

A central methodological distinction governs the engine: country performance and documentary confidence are different measures. A country can have weak performance documented by strong evidence. France illustrates this principle: a performance score of 52/100 can coexist with high documentary confidence because official sources, including critical findings, strongly support the assessment. Conversely, a high provisional performance score supported only by generic or policy-level evidence should not be treated as reliable.

The engine therefore separates four questions:

- What is the proposed performance of the country across the six IML domains?
- How strong and specific is the documentary evidence supporting that proposal?
- Does the evidence describe policy, architecture, implementation, outcomes or correction?
- Has a qualified human reviewer examined, corrected and validated the profile?

Automation may search, organise, compare and flag inconsistencies. It may calculate arithmetic summaries and documentary confidence. It must not convert the presence of official documents into proof of successful implementation, and it must not publish a country profile as validated without human responsibility.

1. Purpose and Relationship to the Founding Manuscript

1.1 Why a separate technical manuscript is necessary

The Founding Manuscript is intentionally broader than a software specification. It defines Health Information Ecosystems, explains interoperability as preservation of meaning, trust, context and usefulness, and presents the six IML assessment domains. It also describes evidence grading, human review, country profiles rather than rankings, database quality and the Open Clinical Workspace. A detailed database schema, audit workflow and API contract would overload that scientific narrative.

This companion document therefore carries the implementation detail. It can evolve more frequently than the Founding Manuscript while remaining subordinate to its principles. Changes to tables, functions, interfaces or deployment technology do not alter the scientific purpose of IML unless they change definitions, indicators, scoring rules or validation requirements.

1.2 Division of responsibility between documents

Founding Manuscript	Technical Manuscript
Scientific purpose, definitions and hypotheses	Database structures, functions, workflows and API contracts
Six domains and cross-cutting dimensions	Storage and calculation of domain scores and evidence
Evidence grading and methodological caution	Operational representation of evidence level, direction, scope and implementation status
Human responsibility and review	Review gates, audit records and publication controls
Open Clinical Workspace vision	Current country-profile engine and future administrative tooling
Roadmap and research agenda	Implementation roadmap, versioning and migration strategy

1.3 Intended readership

This document is written for researchers, clinicians, digital-health experts, database engineers, software contributors, public institutions and reviewers who need to understand how IML country information is produced. It is also intended to make the prototype inspectable by future academic or public-sector partners.

2. Scope, Non-Objectives and Design Principles

2.1 Scope

- Structured storage of country profiles and their versions.
- Storage of the six IML domain scores and the resulting overall score.
- Registration of strengths, points to watch and explanatory subtitles.
- Registration of documentary sources and source-to-indicator evidence links.
- Separation of documentary confidence from country performance.
- Human review, correction, validation and publication status.
- Publication of profile data through a stable API to the public website.
- Preparation for a future administrative interface and reproducible country research workflow.

2.2 Non-objectives

- The engine is not a global country ranking system.- It is not an autonomous AI evaluator and does not provide a definitive national judgement. • The engine is not a global country ranking system.
- It does not infer successful implementation from laws, strategies or national platforms alone. • It is not an autonomous AI evaluator and does not provide a definitive national judgement.
- It does not replace national experts, public authorities, peer review or field validation. It does not infer successful implementation from the existence of laws, strategies or national platforms.- It is not a clinical production database and currently stores no patient-level health data. • It does not replace national experts, public authorities, peer review or field validation. It does not claim that the current provisional country profiles are scientifically validated.

2.3 Design principles

Principle	Operational meaning
Health first	Technical choices are assessed by their ability to support health, not by novelty or market position.
Evidence before assertion	Every important country statement should be traceable to one or more sources, with limitations.
Human responsibility	The engine supports research; a human reviewer remains accountable for interpretation and publication.
Separation of measures	Performance, documentary confidence and validation status remain distinct.
Versioning	Country profiles and methods must be reproducible over time.
Correctability	Errors, contested interpretations and outdated sources must be identifiable and amendable.
Vendor neutrality	The architecture uses open standards and avoids dependency on one software provider.
Progressive implementation	The system can begin with a small public profile layer and evolve toward richer audit and administrative functions.

3. System Architecture

The prototype is organised into five complementary layers. The layers are intentionally separated so that publication cannot bypass evidence handling and human review.

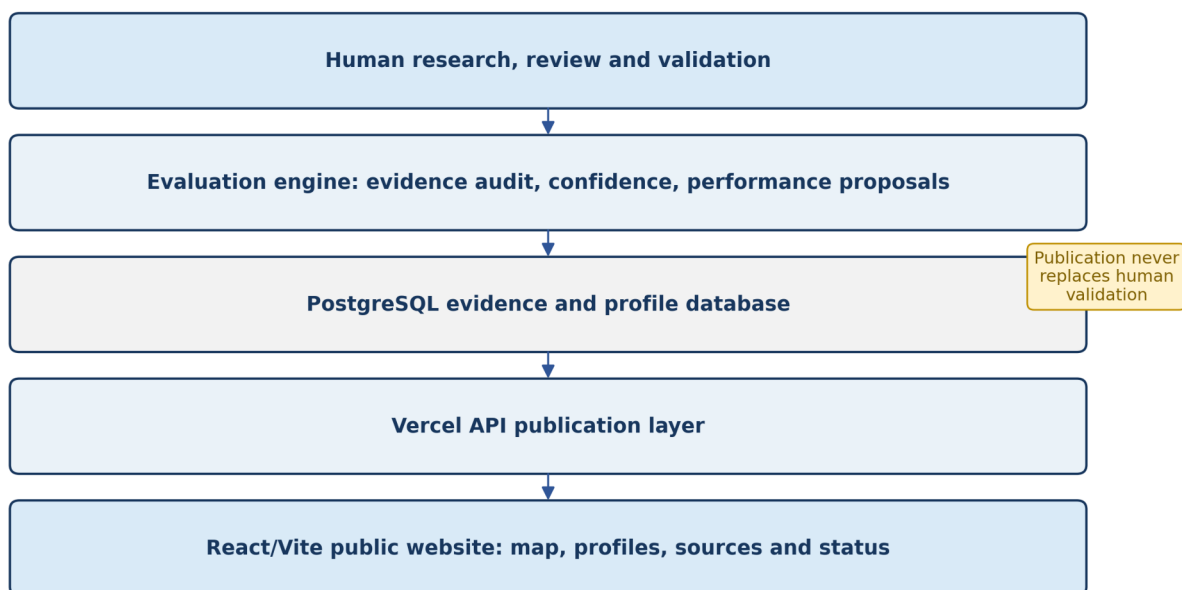


Figure 1. IML country-profile architecture: public publication remains downstream of evidence organisation and human validation.

3.1 Human research and validation layer

Researchers identify and interpret sources, examine implementation reality, investigate contradictions and propose corrections. Reviewers assess whether a profile is sufficiently supported and whether its scores reflect the evidence. This layer is methodologically primary even though it is not a software component.

3.2 Evaluation engine

The engine organises evidence, performs deterministic checks, calculates documentary confidence, compares stored and calculated values, and produces warnings. It can support an assessor by exposing missing domains, generic evidence, contradictory findings, weak implementation evidence and outdated sources.

3.3 PostgreSQL evidence and profile database

PostgreSQL is the system of record. It preserves country metadata, profile versions, domain scores, notes, sources, evidence links, audit results and assessment decisions. Referential constraints prevent orphaned records and make each published profile traceable to its evidence.

3.4 API publication layer

The Vercel serverless API reads published profiles from PostgreSQL and returns structured JSON. The current API route selects active countries with a published profile, calculates the overall score from the six domain scores, and returns notes, sources, indicator links and the latest assessment record.

3.5 Public website

The React/Vite website presents the World Map and Country Report. It is a communication and review interface, not the authoritative research workspace. The site should display profile status, source limitations and validation state clearly enough to prevent exploratory information from appearing definitive.

4. PostgreSQL Data Model

4.1 Two complementary data zones

Publication zone and audit zone

The database separates information consumed by the public website from the working audit process. Publication tables contain the current profile representation. Audit tables and functions document how sources were checked and how evidence was interpreted. This separation allows a rigorous workflow without forcing the public API to expose every internal research record.

4.2 Core publication tables

Table	Role	Main content
countries	Country identity and map presentation	ISO3 code, slug, English name, map coordinates, label offsets, active flag
country_profiles	Versioned country profile	Country reference, version, status, subtitle, assessment date, publication date
iml_domains	Reference list of six domains	Code, title, display order
country_profile_scores	Performance by domain	Profile, domain code, score 0–100
country_profile_notes	Concise interpretation	Strength or watch note, display order, text
country_profile_sources	Documentary source register	Title, publisher, URL, publication/access dates, evidence note
country_profile_source_indicators	Evidence link	Source, indicator code, evidence level, support type, summary, limitation
country_profile_assessments	Human review record	Status, method, reviewer identity/role, confidence, notes, dates, optional overall score

4.3 Audit and quality-control tables

Component	Purpose	Examples
country_profile_source_audits	One audit record for a source or source/profile relationship	Audit status, timestamps, calculated documentary reliability and reviewer metadata
country_profile_source_audit_checks	Individual reproducible checks	Check code, value, comment, completion status
Quality audit view	Country-level quality summary	Performance score, documentary confidence, evidence balance, domain coverage and warnings

The audit tables are designed as a working production layer. The public profile tables are designed as a stable publication layer. An audit may be revised many times before a corresponding profile version is published.

4.4 Key relationships

- One country can have several profile versions.
- One profile has up to six domain score records, one for each IML domain.
- One profile can have many strengths, watch notes and documentary sources.
- One source can support, partially support, contradict or contextualise several indicators.
- One profile can have several assessment records, preserving the history of review.
- Only a profile explicitly marked as published should be returned by the public API.

4.5 Integrity constraints

- ISO3 country codes should be unique.
- The pair country_id + version should be unique for profiles.
- The pair profile_id + domain_code should be unique for scores.
- Scores should remain within 0–100.
- Evidence level and status values should be controlled by check constraints or reference tables.
- Deletion rules should avoid accidental loss of published evidence and should prefer explicit archival or supersession.
- Every published profile should have six domain scores, a minimum source set and a human assessment status.

5. Country Profile Lifecycle

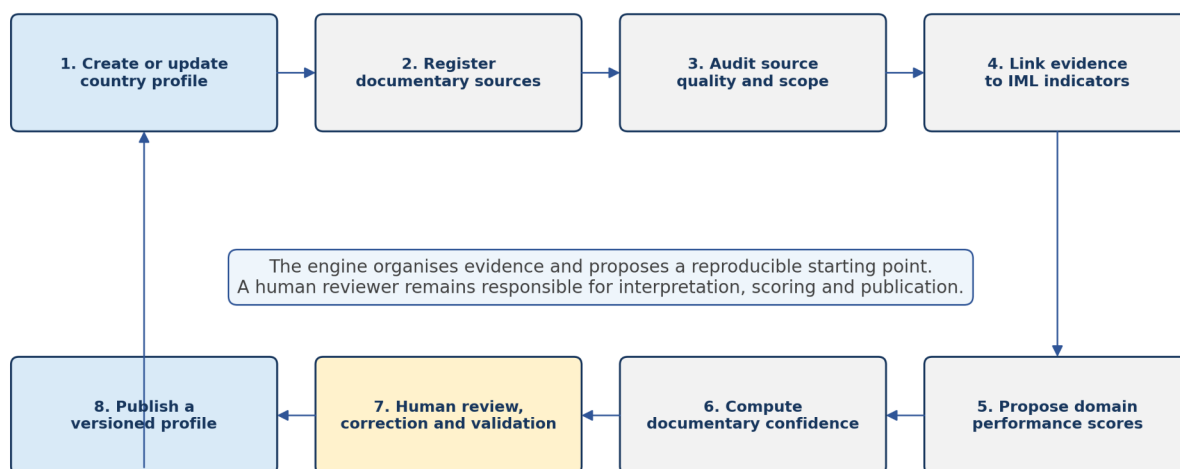


Figure 2. Country profile lifecycle. Automation prepares and checks the record; human review governs interpretation and publication.

5.1 Working states

A country profile should move through explicit states rather than appearing directly as published. The exact vocabulary can evolve, but the underlying lifecycle should remain stable.

State	Meaning	Public visibility
draft	Initial profile under construction	No
proposed	Scores and evidence assembled for review	Normally no
reviewed	Human reviewer has examined the profile; corrections may remain	Optional, clearly labelled
validated	Methodological and country review requirements satisfied	Yes
published	Version released to the public API	Yes
superseded	A newer version replaces the record	Historical only

5.2 Current prototype simplification

The current public application uses a simpler profile status, with published profiles returned by the API. This is sufficient for the prototype but does not yet capture the full scientific review lifecycle. The technical roadmap therefore proposes a richer set of states and a publication gate linked to the latest assessment record.

5.3 Versioning rule

A material change to evidence, methodology or domain scores should create a new profile version rather than silently rewriting the historical record. Minor corrections to typographical errors or broken links may be logged without changing the score version, provided that the correction does not alter interpretation.

6. Evidence Model and Source Audit

6.1 Source registration

A source record identifies the document or official page used in a country assessment. It should include a stable title, publisher, URL, publication date when available, access date and a concise note explaining why the source is relevant. A source is not evidence until it is linked to one or more IML indicators with an explicit interpretation.

6.2 Evidence levels

Level	Typical meaning	Caution
A	Strong primary, official, audited or high-quality evidence directly relevant to the indicator	Level A can document success or failure; it does not imply a positive result.
B	Good evidence that partially covers the indicator, implementation or national scope	Important limitations must be explicit.
C	Indirect, incomplete, regional, secondary or older evidence	Useful for triangulation, not sufficient alone for strong claims.
D	Contextual, indicative or seed evidence	Suitable for starting research, not for validating a score.

6.3 Support type

Value	Meaning
supports	The source directly supports the documented finding.
partially_supports	The source supports only part of the finding, population, territory or time period.
contradicts	The source provides evidence against a proposed assertion or score.
context_only	The source provides background but does not validate the finding.

6.4 Evidence direction

Evidence level describes strength; evidence direction describes the orientation of the documented finding. The distinction is essential. A critical official audit can be high-quality negative evidence, while a national strategy can be high-quality evidence that a policy exists without proving effective implementation.

Direction	Interpretation
positive	The source documents a functioning capability or favourable result.
negative	The source documents a failure, gap, delay, inequity, weakness or ineffective correction.
mixed	The source documents both capability and important limitations.
neutral	The source is descriptive and does not support a directional judgement.
unknown	Direction has not yet been assessed and should block confident publication.

6.5 Evidence scope

Scope	Question answered
governance_policy	Is there a law, strategy, mandate, responsible institution or governance process?
architecture	Is there a technical, semantic, identity or security architecture?
implementation	Is the capability deployed and used in real organisations and workflows?
outcomes	Does the capability produce measurable clinical, public-health or system results?
correction	Can errors, failures and recommendations be corrected and propagated through the ecosystem?
unknown	The source has not yet been classified.

6.6 Implementation status

Status	Meaning
planned	Announced or legislated but not yet available.
available	Technically or legally available, with insufficient proof of routine use.
operational	Evidence of routine use in the relevant scope.
uneven	Implementation varies by territory, vendor, institution or population.
not_demonstrated	The reviewed evidence does not establish implementation.
not_assessed	No implementation judgement has been recorded.

6.7 Source audit checks

- Authority and provenance: who published the source and under what responsibility?
- Relevance: which IML indicator and domain does it address?
- Timeliness: is the source current enough for the assessed period?
- National scope: does it apply nationally, regionally, institutionally or only to a pilot?
- Implementation: does it document availability, routine use, outcomes or only policy intent?
- Independence: is the source self-description, external audit, research or regulatory oversight?
- Limitations: what cannot be inferred from the source?
- Direction: does the finding support a positive, negative, mixed or neutral interpretation?
- Contradictions: do other sources challenge the finding?
- Reproducibility: can another reviewer retrieve and interpret the same source?

7. Evaluation Engine

7.1 Two separate outputs

Performance is not confidence

The performance score describes the proposed maturity of the country. The documentary confidence score describes how well the proposal is supported. A high confidence score must never increase a country performance score merely because the evidence is strong.

7.2 Country performance score

The current public API calculates the overall score as the arithmetic mean of the six stored domain scores. Each domain score is stored separately and ordered through the `iml_domains` reference table. This simple rule is transparent and reproducible. Any future weighting system must be versioned and documented rather than introduced silently.

The six current domains are:

- Governance and Standards
- Technical Interoperability
- Identity, Consent and Trust
- Adoption and Use
- Security and Resilience
- Feedback, Correction and Learning

7.3 Documentary confidence score

Documentary confidence is an experimental technical metric intended to prioritise review. It is not a scientific result about the country. The current working formula combines evidence strength, domain coverage, number of sources, number of source-to-indicator links, completion of evidence direction and presence of operational evidence.

Component	Experimental weight	Rationale
Evidence level quality	40%	Rewards stronger, more direct evidence.
Coverage of six domains	25%	Avoids confidence based on one narrow area.
Number of sources	10%	Encourages documentary breadth without unlimited volume.
Evidence-link density	10%	Rewards explicit source-to-indicator interpretation.
Direction assessed	10%	Prevents strong evidence from being misread as automatically positive.
Operational evidence	5%	Distinguishes policy or architecture from implementation, outcomes and correction.

The weights are provisional and should be empirically reviewed. The engine should expose the components as well as the total, so that a reviewer can understand why a confidence value is high or low.

7.4 Evidence balance

The engine may summarise the evidence as predominantly positive, predominantly negative, mixed or not assessed. This label is descriptive. It should not directly set the performance score because the significance of each finding depends on the indicator, national context and scope.

7.5 Deterministic quality warnings

- INCOMPLETE_PROFILE: fewer than six domain scores.
- INSUFFICIENT_EVIDENCE: too few sources, links or domains covered.
- DIRECTION_NOT_ASSESSED: one or more evidence links still have unknown direction.
- FRAMEWORK_ONLY: sources document policy or architecture but no implementation, outcome or correction evidence.
- GENERIC_SEED_EVIDENCE: most links remain level D or use generic summaries and limitations.
- READY_FOR_REVIEW: minimum documentary conditions are met, but human validation is still required.

7.6 Current database functions

The working audit branch has tested functions for retrieving the active profile, registering sources, creating source audits, initialising checks, recording check results, analysing documentary quality and validating an audit. These functions support reproducibility by moving repeated logic into PostgreSQL rather than leaving it in ad hoc manual queries.

Function	Role
iml_get_active_profile()	Resolve the current working profile for a country or workflow.
iml_get_sources()	Return the source set linked to a profile.
iml_create_source_audit()	Create an audit record for a source.
iml_initialize_audit_checks()	Create the standard checklist for the audit.
iml_set_audit_check()	Record or update a check result.
iml_analyze_audit()	Calculate documentary reliability and identify missing checks.
iml_validate_audit()	Confirm that the audit satisfies the validation conditions.

7.7 What automation may and may not do

Automation may	Automation may not
Retrieve and organise candidate sources	Declare national implementation from policy documents alone
Detect broken links, missing dates and duplicate sources	Replace a country expert or institutional reviewer
Classify provisional evidence level, direction and scope	Publish a profile as validated without human approval
Calculate transparent arithmetic and confidence metrics	Treat documentary confidence as country performance
Flag contradictions and missing domains	Hide uncertainty or unsupported assumptions
Generate a draft narrative and proposed scores	Present provisional scores as definitive facts

8. Human Research and Validation Protocol

8.1 Researcher responsibilities

- Define the assessment period and country scope.
- Collect official, regulatory, audited, scientific and implementation sources.
- Distinguish national availability from routine use and measurable outcomes.
- Link each source to specific indicators rather than to a country in general.
- Record an evidence summary and a limitation in plain language.
- Classify evidence level, support type, direction, scope and implementation status.
- Investigate contradictions and regional variation.
- Propose domain scores with an explicit rationale.
- Document uncertainties and questions requiring country expertise.

8.2 Reviewer responsibilities

- Confirm that sources are retrievable, relevant and accurately represented.
- Check whether negative evidence has been treated as negative rather than as a positive institutional feature.
- Check whether policy and architecture have been confused with implementation or outcomes.
- Review domain scores against the complete evidence balance.
- Assess whether the profile language is proportionate and avoids national stereotyping.
- Correct errors, request additional evidence or reject unsupported claims.
- Record reviewer name, organisation, role, method, confidence and review notes.
- Approve, defer or refuse publication of the profile version.

8.3 Minimum publication gate

Requirement	Minimum expectation
Profile structure	Six domain scores and an explanatory subtitle
Evidence	Several retrievable sources with indicator-level links
Coverage	Evidence across the principal domains, with gaps explicitly visible
Interpretation	Specific evidence summaries and limitations; no generic seed text
Direction and scope	Completed for all material evidence links
Implementation evidence	At least some evidence beyond policy or architecture
Human review	Named reviewer or documented review process
Status	Validated or explicitly provisional; never silently ambiguous
Versioning	Assessment date, publication date and version preserved

8.4 Country validation

Country validation should ideally involve a national or regional expert, institution or practitioner familiar with the real operation of the health information ecosystem. The expert is not asked merely to endorse the profile. The purpose is to identify factual errors, missing structures, implementation variation, recent reforms and unintended interpretations. Disagreement should be recorded rather than erased.

8.5 Review outcomes

Outcome	Action
Validated	Publish or maintain the version with reviewer metadata.
Validated with limitations	Publish with visible uncertainty and stated gaps.
Revision required	Return to research; do not present as validated.
Insufficient evidence	Reduce claims, lower confidence or suspend the profile.
Contested	Publish only if disagreement and alternative evidence are transparently documented.
Superseded	Retain for history and publish the newer version.

9. Publication API and Public Website

9.1 Current API behaviour

The current countries-v2 API route connects to PostgreSQL using the DATABASE_URL environment variable. It selects active countries joined to published profiles, returns map presentation metadata, domain scores, strengths, points to watch, sources, indicator evidence and the latest assessment record. The response includes an API version, generation timestamp, country count and country array.

9.2 Current public payload

For each country, the API exposes: ISO3 code, slug, name, map coordinates, profile identifier and version, status, subtitle, dates, overall score, six values, domain objects, strengths, watch points, documentary sources, indicator evidence and assessment metadata.

9.3 Publication safeguards

- The API should return only the profile version intended for publication.
- Cache behaviour should not prevent rapid correction of newly published evidence.
- Assessment and confidence metadata should be exposed when the interface can display them clearly.
- Internal audit comments containing personal or sensitive reviewer information should remain outside the public payload.
- Future API versions should preserve backward compatibility or publish a documented migration path.
- The website should not imply that provisional profiles are validated national rankings.

9.4 Administrative interface

A future administrative interface should allow authorised contributors to create a country, edit a profile version, add sources, classify evidence, enter proposed scores, run quality checks and submit the profile for review. Publication should require a separate explicit action and should never occur automatically after data entry.

10. Illustrative Country Contrasts

Illustrative, not comparative ranking

The examples below explain the distinction between performance and documentary confidence. They should not be interpreted as a definitive comparison between countries.

10.1 France: low performance can be strongly documented

The current French profile has an overall performance score of 52/100. Its strengths include identity and security foundations, while technical interoperability, adoption and especially feedback/correction remain weak. A critical Cour des comptes report provides strong evidence of delayed implementation, limited enforceability, uptake below expectations and persistent institutional blockages. This evidence is high-quality and predominantly negative. It increases confidence in the low score; it does not improve the score.

10.2 Germany: stronger performance with mixed implementation evidence

The current German profile has an overall performance score of 81/100. Official sources strongly document the statutory framework, national Telematikinfrastruktur, electronic patient record and security architecture. However, some evidence is mixed because national availability does not by itself prove seamless integration across vendors and care organisations. Human review should therefore preserve the distinction between formal national capability and uniform operational performance.

10.3 Generic seed profiles

Many initial profiles were created with five indicative official sources classified as level D and context_only. These records are useful as a starting map of institutions and themes, but they do not validate the stored scores. The engine should identify such profiles as generic seed evidence and prioritise them for human research rather than silently upgrading them.

Example	Performance score	Documentary confidence	Interpretation
France	52	High in the experimental audit	Weak performance supported by strong positive and negative evidence.
Germany	81	High in the experimental audit	Stronger formal and technical capacity, with implementation limitations still requiring review.
Generic seed profile	May appear high or low	Low to moderate	Score remains exploratory because evidence is generic, incomplete or policy-heavy.

11. Security, Integrity and Reproducibility

11.1 Current data sensitivity

The current country-profile database stores public documentary information and methodological metadata, not patient-level health information. Nevertheless, integrity and access control matter because country profiles can influence institutional interpretation and public trust.

11.2 Security controls

- Environment variables must protect database credentials; credentials must never appear in source code or public outputs.
- Production, preview and development environments should use explicitly managed connections.
- Database roles should follow least privilege, separating read-only publication from administrative writes.
- Administrative actions should be logged with time, actor and profile version.
- Backups and restore procedures should be tested, not merely configured.
- URLs and source metadata should be sanitised before display.
- Future reviewer accounts require role-based access and secure authentication.
- Any later use of non-public or patient-level data would require a separate legal, ethical and security architecture.

11.3 Reproducibility

- Preserve the SQL schema and migrations in version control.
- Version scoring and confidence formulas.
- Preserve source access dates and profile assessment dates.
- Retain superseded profile versions.
- Export machine-readable profile snapshots.
- Document which code revision and database migration produced each public release.
- Use deterministic calculations for overall score and quality warnings.
- Allow another reviewer to reproduce the evidence set and calculations.

11.4 Correctability

Correctability is a first-class requirement. The system should support correction of source metadata, evidence interpretation, scores and narratives without erasing the historical record. A correction should identify what changed, why it changed, who approved it and which public version supersedes the earlier one.

12. Governance and Versioning

12.1 Governance roles

Role	Responsibility
Author or profile researcher	Collect evidence, propose interpretations and draft scores.
Methodological reviewer	Check consistency with IML definitions and scoring rules.
Country or subject-matter reviewer	Check factual and operational accuracy.
Technical maintainer	Maintain schema, migrations, API and deployment.
Publication approver	Authorise release of a profile version.
Public contributor	Submit corrections or additional sources through a transparent process.

12.2 Method versioning

The database schema version, indicator catalogue version, scoring method version and public API version should be recorded independently. A profile can therefore state not only when it was published, but also which methodology and technical schema produced it.

12.3 Change categories

Change	Typical version effect
Editorial correction	No score version change; log the correction.
Source replacement without interpretation change	Update evidence metadata; retain profile version if justified.
New evidence affecting a domain score	Create a new profile version.
New indicator or domain definition	Create a new methodology version and reassess affected profiles.
Confidence formula adjustment	Create a new technical/method version; do not retroactively hide earlier values.
API field change	Increment API version or preserve backward compatibility.

13. Limitations and Research Questions

13.1 Documentary asymmetry

Countries differ in the volume, language, accessibility and transparency of public documentation. A low confidence score may reflect limited public evidence rather than weak national capability. The system must avoid converting documentation visibility into a hidden geopolitical bias.

13.2 National averages conceal variation

Federal, regional and decentralised systems may have substantial internal variation. A single national score can obscure high-performing regions and poorly connected care pathways. Future profiles should be able to record subnational evidence or scope limitations without creating false precision.

13.3 Official self-description and independent evidence

Official sources are indispensable for laws, architecture and national programmes. They are less sufficient for evaluating implementation, outcomes and correction. External audits, scientific studies, professional experience and patient evidence are needed for triangulation.

13.4 Temporal change

Digital-health systems evolve rapidly. Scores and source interpretations can become obsolete. The engine therefore requires assessment dates, source access dates, periodic review and visible supersession.

13.5 Scoring uncertainty

A numeric score can create an impression of precision exceeding the evidence. IML should continue to develop score rationales, confidence intervals or maturity bands, domain-specific narratives and explicit uncertainty rather than relying on a single overall number.

13.6 Research questions

- What minimum evidence set is sufficient for a credible country profile?
- How should policy, architecture, implementation, outcomes and correction be weighted?
- How can country experts participate without turning validation into institutional self-certification?
- How should contradictory evidence affect domain scores and documentary confidence?
- Can the confidence model be calibrated against independent expert review?
- How should subnational variation be represented?
- Which parts of source classification can be automated safely?
- How can public correction requests be integrated without compromising governance?

14. Development Roadmap

14.1 Immediate consolidation

- Complete the separation of performance score and documentary confidence in the database and interface.
- Expose evidence direction, scope and implementation status in the internal workflow.
- Confirm the country-level quality audit view and test it against France and Germany.
- Add visible assessment status to the public Country Report.
- Preserve the current map and profile interface while technical migration is tested.
- Document all current schema objects and migrations in the repository.

14.2 Country research workflow

- Upgrade generic level-D source sets country by country.
- Prioritise profiles with high scores but weak implementation evidence.
- Create a structured review sheet for national experts.
- Add contradiction handling and evidence recency checks.
- Create profile snapshots before and after human validation.

14.3 Administrative application

- Create authenticated forms for countries, profiles, scores, notes and sources.
- Add source audit checklists and automated warnings.
- Add reviewer assignment and review decisions.
- Add explicit publish, unpublish and supersede actions.
- Provide export to JSON, CSV and a human-readable country dossier.

14.4 Scientific validation

- Publish the technical methodology for external review.
- Compare engine output with independent expert scoring.
- Test inter-rater agreement and revise indicator guidance.
- Validate the documentary confidence formula empirically.
- Create a governance model for academic or public-sector stewardship.

Annex A. Data Dictionary

A.1 countries

Field	Purpose
id	Internal primary key.
iso3	Unique ISO-style three-letter country code used by the application.
slug	Stable URL and application identifier.
name_en	English display name.
map_x / map_y	Map position in the current World Map coordinate system.
label_dx / label_dy	Country label offsets.
text_size	Map label size.
is_active	Controls whether the country is eligible for publication.
created_at / updated_at	Record lifecycle metadata where present.

A.2 country_profiles

Field	Purpose
id	Profile primary key.
country_id	Reference to countries.
version	Country-specific profile version.
status	Working/publication state.
subtitle	Concise country interpretation.
assessment_date	Date to which the assessment refers.
published_at	Public release timestamp.
created_at / updated_at	Lifecycle metadata.

A.3 country_profile_scores

Field	Purpose
profile_id	Reference to the profile.
domain_code	Reference to one of the six IML domains.
score	Proposed country performance value from 0 to 100.

A.4 country_profile_notes

Field	Purpose
profile_id	Reference to the profile.
note_type	Strength or watch.
display_order	Stable display ordering.
note_text	Concise interpretive statement.

A.5 country_profile_sources

Field	Purpose
profile_id	Profile to which the source belongs.
title	Document or page title.
publisher	Responsible organisation.
source_url	Retrievable location.
publication_date	Original publication date when known.
accessed_at	Date reviewed by IML.
evidence_note	Why the source is relevant to the profile.

A.6 country_profile_source_indicators

Field	Purpose
source_id	Reference to the documentary source.
indicator_code	IML indicator addressed.
evidence_level	A, B, C or D documentary strength.
support_type	supports, partially_supports, contradicts or context_only.
evidence_summary	Specific finding supported by the source.
limitation_note	What cannot be inferred.
evidence_direction	positive, negative, mixed, neutral or unknown.
evidence_scope	policy, architecture, implementation, outcomes, correction or unknown.
implementation_status	planned, available, operational, uneven, not_demonstrated or not_assessed.

A.7 country_profile_assessments

Field	Purpose
profile_id	Reviewed profile.
assessment_status	Draft, proposed, reviewed, validated, published or other controlled state.
assessment_method	Description or version of the review method.
reviewer_name	Named reviewer when appropriate.
reviewer_organisation	Institutional affiliation.
reviewer_role	Country expert, methodological reviewer, technical reviewer or other role.
confidence_level	Reviewer confidence, distinct from calculated documentary confidence.
review_notes	Corrections, limitations and rationale.
reviewed_at / published_at	Decision timestamps.
overall_score	Optional reviewed overall score when the model stores it explicitly.

Annex B. Automated Quality Checks

Check code	Purpose
PROFILE_6_DOMAINS	The profile contains exactly one score for each of the six IML domains.
SOURCE_MINIMUM	The profile has a minimum documentary source set.
DOMAIN_COVERAGE	Evidence links cover several domains rather than one narrow theme.

Check code	Purpose
NO_GENERIC_SUMMARY	Evidence summaries are specific and do not use seed placeholders.
NO_GENERIC_LIMITATION	Limitations are specific to the source and claim.
DIRECTION_COMPLETE	Material evidence links have assessed direction.
SCOPE_COMPLETE	Material evidence links have assessed scope.
IMPLEMENTATION_EVIDENCE	At least some evidence addresses implementation, outcomes or correction.
SOURCE_DATE	Publication and access dates are recorded where possible.
URL_RETRIEVABLE	The source URL is retrievable or an archived copy is recorded.
CONTRADICTION_REVIEW	Contradictory evidence has been acknowledged.
ASSESSMENT_PRESENT	A human assessment record exists.
PUBLICATION_GATE	The profile meets the configured requirements before publication.
VERSION_METADATA	Method, profile and API versions are recorded.

Annex C. Profile and Evidence Statuses

C.1 Profile lifecycle statuses

Status	Meaning
draft	Research record under construction.
proposed	Complete enough for formal review.
reviewed	Reviewed but not necessarily validated.
validated	Human validation requirements satisfied.
published	Released through the public API.
superseded	Replaced by a newer version.
withdrawn	Removed from publication because of material concerns.

C.2 Documentation quality statuses

Status	Meaning
INCOMPLETE_PROFILE	Structural data are missing.
INSUFFICIENT_EVIDENCE	Source volume or domain coverage is inadequate.
GENERIC_SEED_EVIDENCE	Sources are primarily level D and context_only.
DIRECTION_NOT_ASSESSED	Evidence strength exists but its orientation is unknown.
FRAMEWORK_ONLY	Only policy or architecture is documented.
PROVISIONAL	Evidence is partly developed but not review-ready.
READY_FOR_REVIEW	Minimum automated conditions are satisfied; human validation remains pending.

Annex D. Human Publication Checklist

- The country and assessment period are correctly identified.
- All six domain scores are present and justified.
- The overall score matches the configured calculation method.
- Each material claim is traceable to one or more sources.
- Evidence level, support type, direction, scope and implementation status are complete.
- Negative evidence has not been misinterpreted as positive capacity.
- Policy or architecture has not been presented as proof of routine implementation.
- Regional and institutional variation is acknowledged.
- Contradictory evidence is documented.
- Strengths and watch points are proportionate to the evidence.
- The profile does not imply a definitive national ranking.

- Uncertainty and missing evidence are visible.
- A reviewer has recorded identity or review role, method, date and notes.
- The publication status and profile version are correct.
- The API and website display the intended version.
- A supersession or correction route exists.

Annex E. Example API Payload

The following simplified example illustrates the public structure. It omits internal audit records and uses placeholder values.

```
{
  "api_version": "2",
  "generated_at": "2026-07-18T00:00:00.000Z",
  "count": 1,
  "countries": [
    {
      "iso3": "XXX",
      "slug": "example-country",
      "name": "Example Country",
      "profile_id": "1",
      "version": 1,
      "status": "published",
      "subtitle": "Illustrative profile text.",
      "overall_score": 60,
      "values": [60, 55, 70, 50, 75, 50],
      "domains": [
        { "code": "governance", "score": 60, "display_order": 1 }
      ],
      "strengths": ["Example strength"],
      "watch": ["Example point to watch"],
      "sources": [
        {
          "title": "Example source",
          "publisher": "Example institution",
          "url": "https://example.org/source",
          "indicators": [
            {
              "code": "GOV-1",
              "evidence_level": "B",
              "support_type": "partially_supports",
              "summary": "Specific documented finding.",
              "limitation": "Specific limitation."
            }
          ]
        }
      ],
      "assessment": {
        "assessment_status": "validated",
        "assessment_method": "IML country review protocol",
        "confidence_level": "moderate"
      }
    }
  ]
}
```

Annex F. Suggested Insertions in the Founding Manuscript

The following short passages are designed for manual insertion into the current Founding Manuscript. They keep the conceptual manuscript readable while directing technical detail to this companion document.

Front matter or How to Read this Manuscript

Technical implementation note. The database architecture, evidence-audit workflow, country-profile lifecycle, publication API and evaluation engine are developed in a separate companion document: IML Database and Country Evaluation Engine: Technical Manuscript for Evidence-Guided, Human-Validated Country Profiles. The technical document supports implementation and reproducibility but does not replace the scientific definitions, methodological caution or human validation requirements established in this Founding Manuscript.

Part V - Methodology, after the sections on evidence grading and validation

Technical implementation of country profiles. IML country profiles may be supported by a structured database and evaluation engine that registers sources, links evidence to indicators, records evidence level, direction, scope and implementation status, and calculates reproducible quality warnings. The engine is intended to provide a documented starting point for research. It does not independently validate a country profile. Interpretation, scoring, correction and publication remain human responsibilities and should be recorded through a versioned review process.

Part V - Methodology, after scoring philosophy

Performance and documentary confidence. IML distinguishes the proposed performance of a Health Information Ecosystem from the confidence of the evidence supporting that proposal. Strong evidence may document weak performance, while a high provisional score may remain unreliable when it is supported only by policy statements or generic sources. Documentary confidence must therefore never be used as a positive component of the country performance score.

Part VIII - Open Clinical Workspace, after Database Quality

Companion technical architecture. The operational database and country evaluation engine are documented separately so that schema design, source audits, API contracts, security controls, versioning and migration can evolve without overloading the Founding Manuscript. The technical architecture should remain vendor-neutral, auditable, correctable and aligned with the health-first principles of IML.

Part IX - Responsible Artificial Intelligence and Scientific Integrity

AI-assisted country research. Artificial intelligence may help identify candidate sources, organise evidence, detect omissions, classify provisional evidence and generate draft summaries. It must not infer successful national implementation from the existence of laws or platforms, replace country expertise, or publish a profile as validated. All country interpretations and final scores remain subject to human review and responsibility.

Part X - Roadmap and Future Work

Technical companion document. The IML roadmap includes a separate Technical Manuscript describing the PostgreSQL database, evidence-audit model, evaluation engine, human review workflow, API publication layer, security and reproducibility requirements. This document should evolve alongside the application and should be externally reviewed before the engine is used for validated country publication.

Change Log

Current working draft - technical companion separation added. Clarified that detailed database architecture, evidence-audit functions, country-profile workflow, documentary-confidence calculations, API publication and migration are developed in a separate IML Technical Manuscript. Added an explicit distinction between country performance and documentary confidence and reaffirmed that automated assistance provides a starting point for human research rather than autonomous country validation.

Implementation Basis Reviewed for this Draft

This technical manuscript was prepared from the current IML Founding Manuscript and the July 2026 prototype implementation, including the PostgreSQL country-profile tables, seed and correction scripts, the countries-v2 serverless API, the Germany profile insertion workflow, the France evidence correction, and the experimental distinction between performance and documentary confidence.

Public website:

Final methodological safeguard

A database can preserve evidence, calculate transparent indicators and expose uncertainty. It cannot replace the situated knowledge of clinicians, institutions, patients, researchers and national experts. IML should use automation to make human judgement more informed, traceable and correctable, not less necessary.

Annex G. Future Universal Health and Research Number

Status. This annex specifies a target architecture for research and pilot evaluation. It does not describe an operational IML identity service, a replacement for national identifiers or an already adopted international standard.

G.1 Design objective

The objective is to create a stable person-level reference capable of recognising the same individual across authorised centres, studies and longitudinal datasets. It should support duplicate detection, correction, clinical and laboratory linkage, biobank and imaging linkage, public-health studies and epidemiological follow-up.

G.2 National source namespaces

The architecture accepts authorised national sources rather than imposing one national model. Candidate namespaces include FRA-INS for the French Identite Nationale de Sante, SWE-PNR for the Swedish personnummer, SWE-SAMORD for a Swedish coordination number when legally appropriate, and DEU-KVNR for the German Krankenversicherthenummer. Additional namespaces require documented legal authority, canonicalisation and governance.

Raw national identifiers should normally remain within the authorised identity or linkage service. The central research dataset should receive only the universal number or a study-specific alias approved for that purpose.

G.3 Illustrative visible structure

<ISO3><SOURCE><GEO><DIGEST>

Examples: FRA-INS-GEO-HASH; DEU-KVNR-GEO-HASH; SWE-PNR-GEO-HASH. These examples are explanatory placeholders. Production syntax requires collision, usability, accessibility and legal testing.

The permanent number should not include IML1, IML2 or another software-version prefix. Algorithm version, key version, issuer, creation time, status and supersession history belong in protected metadata.

G.4 Canonicalisation and keyed derivation

A plain unkeyed hash of a national identifier should not be used because many national identifiers have predictable formats and limited search spaces. The target design should evaluate a keyed construction such as HMAC-SHA-256.

Illustrative canonical input: **ISO3 | SOURCE | NORMALISED_NATIONAL_IDENTIFIER |**

GOVERNED_GEO_REFERENCE. Canonicalisation rules must define character encoding, case, separators, leading zeros, date representation, correction and null handling. The digest may be encoded in Base32 and truncated only after collision analysis.

Secret keys must remain in a hardened server-side key-management environment. They must never appear in client code, study exports, ordinary logs or public repositories. Routine key rotation must not silently change the permanent person-level number. Compromise procedures, re-issuance and reconciliation require explicit governance.

G.5 Personal data status

The universal number, its mappings and the protected digest remain personal data when authorised re-identification or linkage is possible. The system must not describe pseudonymisation as anonymisation. National identity traits may be retained in a protected identity record only where lawfully permitted and necessary for correction or reconciliation.

G.6 Stable number and study-specific aliases

The trusted linkage service may retain the stable universal number while investigators receive a programme-level or study-specific alias. An alias can be generated through a second keyed derivation using the universal number and a study namespace. This limits uncontrolled comparison between datasets while preserving authorised linkage.

G.7 Tokens, QR carriers and GDHCN compatibility

The universal number is not an access token. Access tokens should be temporary, signed, purpose-limited, auditable and revocable. A QR code, wallet or mobile application should carry only such a token or a verifiable digital health certificate, never the national identifier, universal number or health information in clear text.

WHO GDHCN trust specifications may support verification of digital health documents and certificates through public-key infrastructure. Integration should verify document provenance and issuer trust without turning GDHCN into a person registry or exposing the stable research number.

G.8 Multicentre issuance and linkage workflow

1. A participating centre verifies the local or national identity under its legal authority. 2. The authorised service normalises the source identifier and approved geographic reference. 3. A protected linkage service resolves an existing universal number or issues a new one. 4. The local mapping is stored in a protected table. 5. The study receives only the identifier authorised by governance. 6. Corrections, merges and splits propagate through a versioned audit process.

G.9 FHIR and CDISC representation

FHIR should represent the universal number as an **Identifier** with a governed system URI and value.

ResearchSubject.identifier may carry a study participant business identifier. These identifiers remain distinct from the logical id of a FHIR resource. CDISC USUBJID may carry the universal number or a programme-level alias according to the protocol and governance model.

G.10 Error correction and identity safety

The service must manage duplicate universal numbers for one person, incorrect merges of different persons, source-identifier corrections, cryptographic collisions and cross-border conflicts. Every merge, split, supersession and correction must preserve provenance, identify the approving reviewer and trigger downstream correction procedures.

G.11 Governance and pilot phases

Phase 1: specification, source-country analysis and legal review. Phase 2: synthetic-data prototype. Phase 3: authorised single-country pilot. Phase 4: national multicentre pilot. Phase 5: international pilot across contrasting identity systems. Phase 6: external consultation and possible standardisation. No patient-level deployment should precede ethics, privacy, cybersecurity, equity and institutional governance review.

G.12 Reference basis

World Bank ID4D and the Principles on Identification for Sustainable Development; WHO Global Digital Health Certification Network and WHO SMART Trust; French ANS Referentiel INS; Swedish Tax Agency guidance on personal identity and coordination numbers; German health-law and BMG material on the KVNR; HL7 FHIR R5 Identifier and ResearchSubject; CDISC SDTMIG; IETF RFC 2104 on HMAC; applicable data-protection and health-data law.

Annex H. Semantic Documentary Validation and Country Source Audit

H.1 Purpose

The country engine is not a web-link collector. Its documentary task is to identify the institutional page, legal text, programme document, registry, implementation report or audit that actually corresponds to an IML indicator. Automated retrieval prepares the inquiry; a human reviewer validates meaning, scope and publication status.

The Global Digital Health Monitor map may provide an initial orientation to country capacity. It does not replace indicator-specific source research, implementation evidence or national expert review.

H.2 Preferred source hierarchy

1. Exact official page or document for the indicator. 2. Official implementation, registry, programme, standard or legal page. 3. Official institutional overview when no precise page exists. 4. Independent audit or peer-reviewed implementation study. 5. Secondary explanatory source for triangulation only. A generic ministry or agency homepage should not be retained when a precise mission, programme, registry, policy or legal page is available.

H.3 Semantic match requirements

Each source must be checked against the claimed title, publisher, jurisdiction, indicator, domain, population, territory, implementation period and evidence scope. A technically reachable page is not valid when it does not support the stated claim. A document may be authoritative yet only contextual if it describes policy without implementation.

H.4 Retrieval-status taxonomy

valid: page retrieved and semantically relevant. **redirected_valid:** official destination changed but remains relevant. **relocated:** original URL is obsolete and a precise replacement was identified. **dead:** persistent 404/410 or confirmed removal with no equivalent located. **blocked_cloudfront:** access denied or challenge generated by CloudFront. **blocked_cloudflare:** Cloudflare challenge or denial. **anti_bot_or_js:** page requires browser execution, session or anti-automation challenge. **transient_failure:** timeout, temporary 5xx or network error. **unresolved:** status cannot yet be determined.

H.5 A blocked page is not a dead source

CloudFront, Cloudflare, geofencing, JavaScript challenges and anti-bot controls can prevent automated retrieval while the page remains available to a human browser. The engine must record the access condition separately from semantic validity. It must not replace a blocked official source with an unrelated homepage merely to obtain a successful HTTP response.

H.6 Automated checks

Automation may record HTTP status, final URL, redirect chain, canonical URL, title, publisher domain, access time, content type, duplicate fingerprint, publication date when detected, and obvious keyword alignment. It may flag homepage-level links, title mismatch, missing dates, repeated sources, outdated evidence, weak national scope and unavailable implementation evidence.

Automation may not decide that a country capability is operational, infer success from a law or platform announcement, classify a politically sensitive finding as definitive, or publish a source replacement without human semantic review.

H.7 Human audit questions

Does the precise page correspond to the stated title and indicator? Is the publisher responsible for the programme or merely commenting on it? Is the evidence national, regional, institutional or pilot-level? Does it document policy, architecture, availability, routine use, outcomes or correction? What cannot be inferred? Are there contradictory sources? Is the link dead, relocated or merely blocked? Should the public profile display the source, a replacement, or a visible limitation?

H.8 Proposed data fields

source_id; claimed_title; observed_title; publisher; original_url; final_url; canonical_url; retrieval_status; http_status; block_type; content_type; publication_date; access_date; last_checked_at; jurisdiction_scope; evidence_scope; implementation_status; semantic_match; replacement_source_id; reviewer_id; reviewer_decision; limitation; audit_version.

H.9 Deterministic warnings

GENERIC_HOMEPAGE; TITLE_MISMATCH; INDICATOR_MISMATCH; DEAD_LINK; RELOCATED_SOURCE; BLOCKED_NOT_DEAD; NO_IMPLEMENTATION_EVIDENCE; NATIONAL_SCOPE_UNCLEAR; OUTDATED_SOURCE; DUPLICATE_SOURCE; CONTRADICTORY_EVIDENCE; HUMAN_REVIEW_REQUIRED.

H.10 Publication rule

A public country profile should publish only sources that have passed semantic review or are clearly labelled as provisional. Every replacement must preserve the original URL, reason for change, reviewer and date. Uncertainty should remain visible. The final country interpretation and score remain human responsibilities.